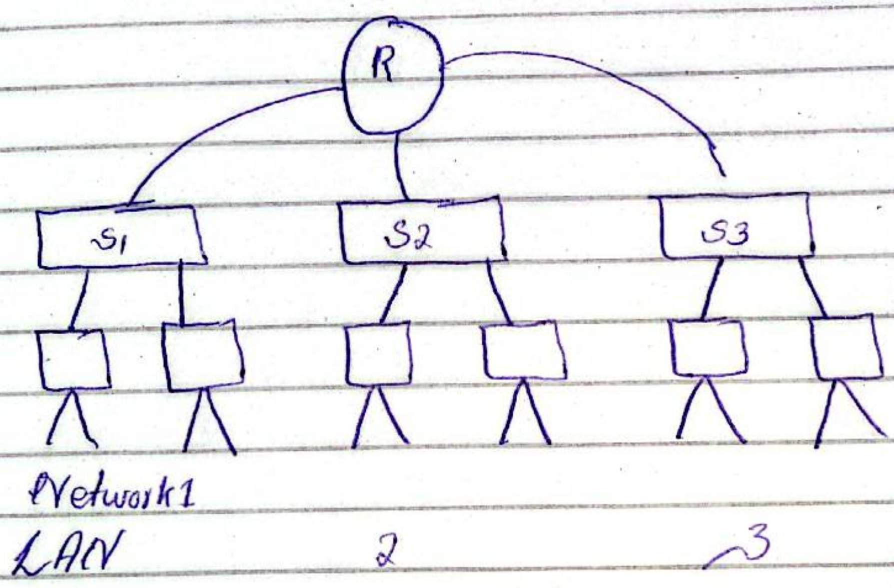
VLANs- Virtual Local area Network.
it is an custom network, which is created from Local area Network. Means from one or more than one local Area Network.

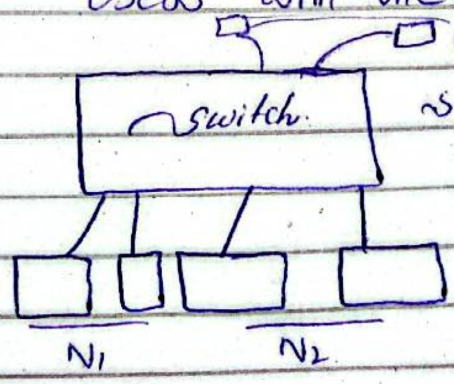
→ let assume we have 3 departments.



So for communication all three ~~routers~~ sw we will need routers.
But this will increase the cost.

So what we will do, we will take only one switch and then we will connect all these users with one switch.

lik this:

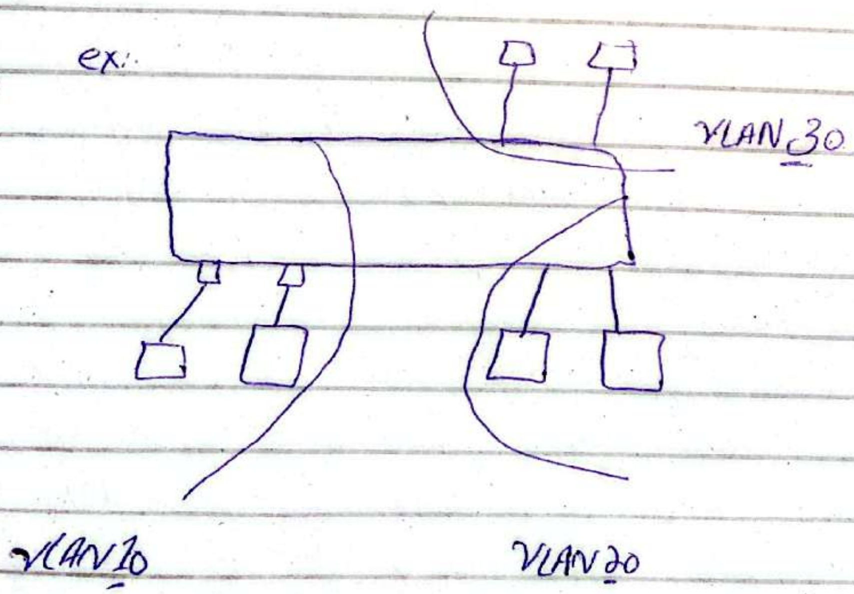


So this is called Virtual area Network.

Note: By default switches all ports exist on one broadcast domain.
so whenever it will generate data, then it will be sent to all.
as dept are differents so it shouldn't be like this..
so for that we will create 3 VLANs. and for each we will stay different different users..
every user for specific department

Note: * VLAN divides the broadcast domains. if we have created VLAN in 3 then it will divide in 3 parts, or if it is in 4 then it will divide in 4 and so on.

ex:



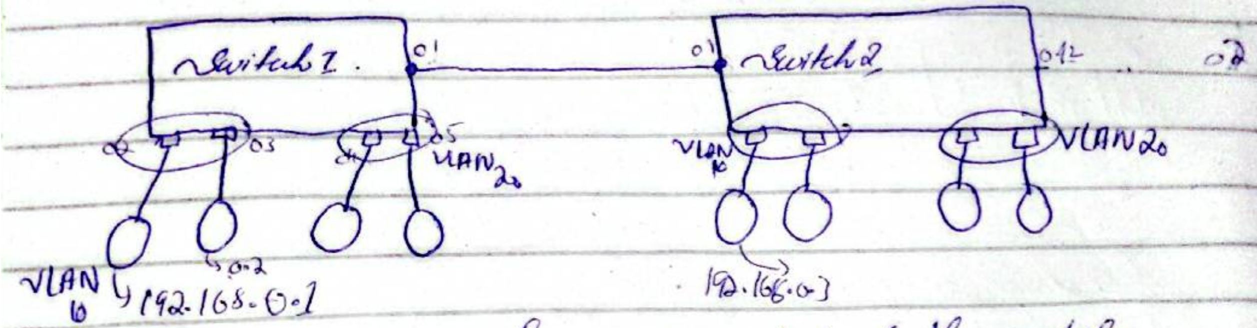
So we want this that our Network 1 will not communicate with 2 or 3.
But these are communicate with the switch and it will broadcast to all users.

So every port we will divide into different VLAN that mean on above switch there is 3 broadcast domain.
So we can create Multiple Broadcast domain in one switch.

So for each VLAN we will give Number.
* So whenever user will generate data then it will go to the switch.
and switch will look that what I have recieved, from which port I have recieved?
like port No1, then it will check port No1, which VLAN Member is?
so if it is the member of VLAN 10,
so it will just talk with VLAN 10 members ports.

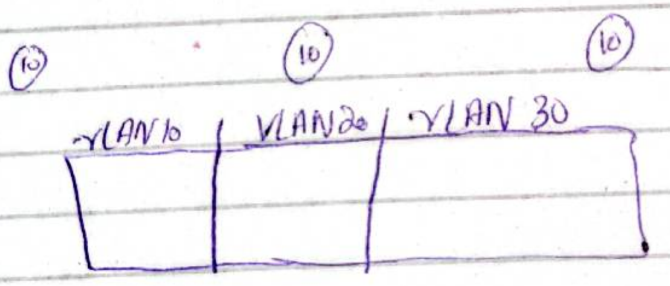
→ trunk Port

So now suppose that we have 2 switches.



Now we have connected both switches and that Port is called trunk port.

∴ So suppose in a company there are 30 users and for thirty users we have 3 departments.



instead of using Router we will take one switch and then we will divide it into VLANs.

So we use the VLAN to Break Broadcast domain.

So Suppose that for each port switch has the Separate Broadcast Domain like Router.

then there were no problem will arrised.
But switch's all ports are divided into one broadcast domain.

So that why we use VLAN to break the broadcast domain.

Practicalay

Switch
> en
> show VLAN ->
> config t
> Vlan 10 // created VLAN 10.
> Name HR // Giving Name.
> VLAN 20 // another VLAN
> Name IT // giving Name.
> int fast ethernet 0/1
> switch port mode Access // assign mode access.
> switch port Access VLAN 10 // dedicating access to VLAN 10
> exit.
> int FastEthernet 0/2
> switchport mode accen
> swithport access VLAN 10.
> do show VLAN brief.

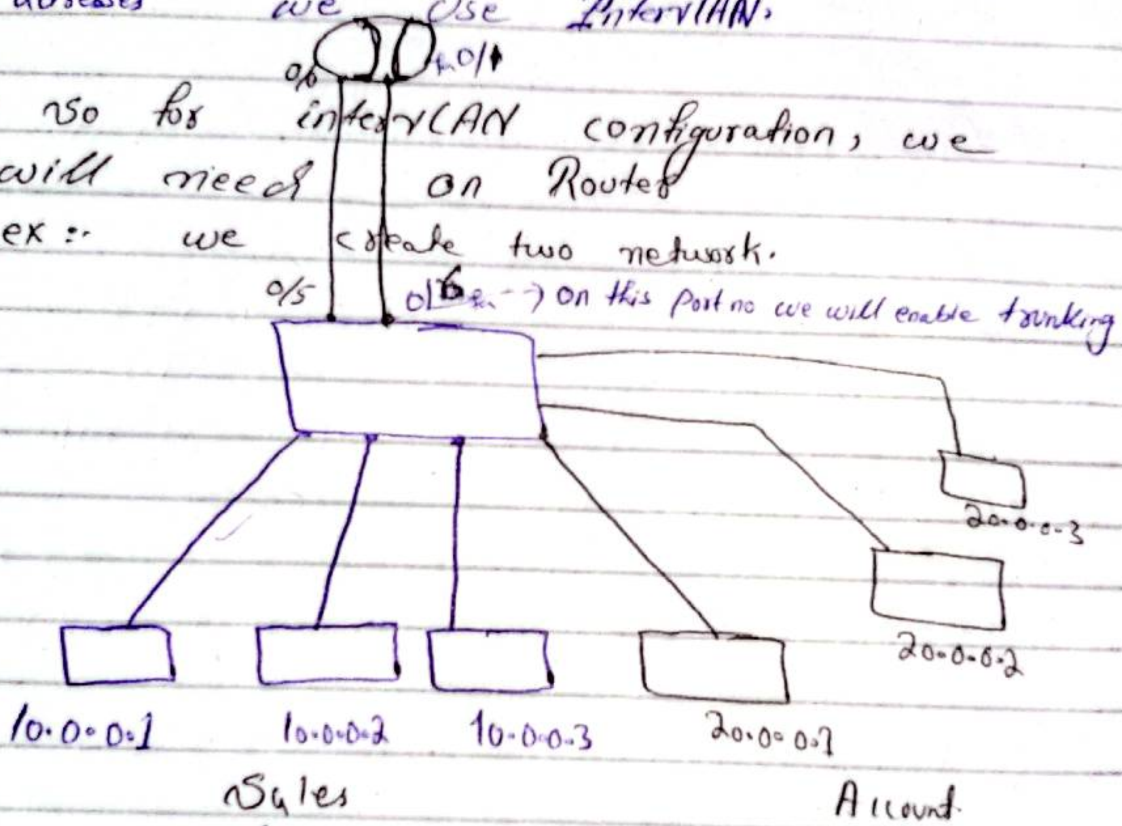
⇒ Inter VLAN Routing :-

→ Inter VLAN Routing we use specially for communicate with the different VLAN along with different Network Ip adresses.

So for these Network which are different ip adresses we use InterVLAN.

⇒ So for interVLAN configuration, we will need on Router

ex :- we create two network.



So we have two different Networks. and we will also use the router.

X

→ so step!

enable the port 0/24 for trunking enable.
and this port is called trunk port

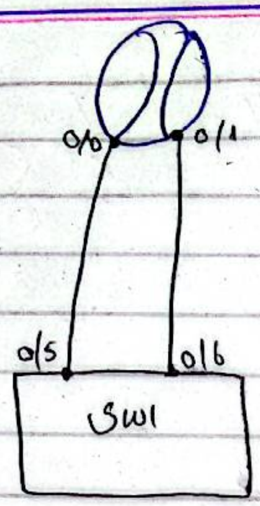
so for this go to the switch.

In cli
config t
Interface fastethernet 0/24
Switchport mode trunk // this port is used for enabling the trunking.

control z.

→ Now Go to router click on cli.
no
enable
show ip interface brief.
configure t
interface fastEthernet 0/0
no shutdown / Enter
for checking step up or not
show ip interface brief
it will show 0/0 Yes unset up.

⇒



⇒ Aftter Assigning the Ip and Gateway.
click on Switch.

en

→ config t

Interface fastethernet 0/5

switchport mode access ~~at~~
switchport access vlan 10

same for 0/6.

Interface fastethernet 0/6
switchport mode access
Switchport access vlan 20

exit.

vtp 22

In Router cli.

en

config t

Int fast 0/0

Ip add 192.168.0.100 255.255.255.0

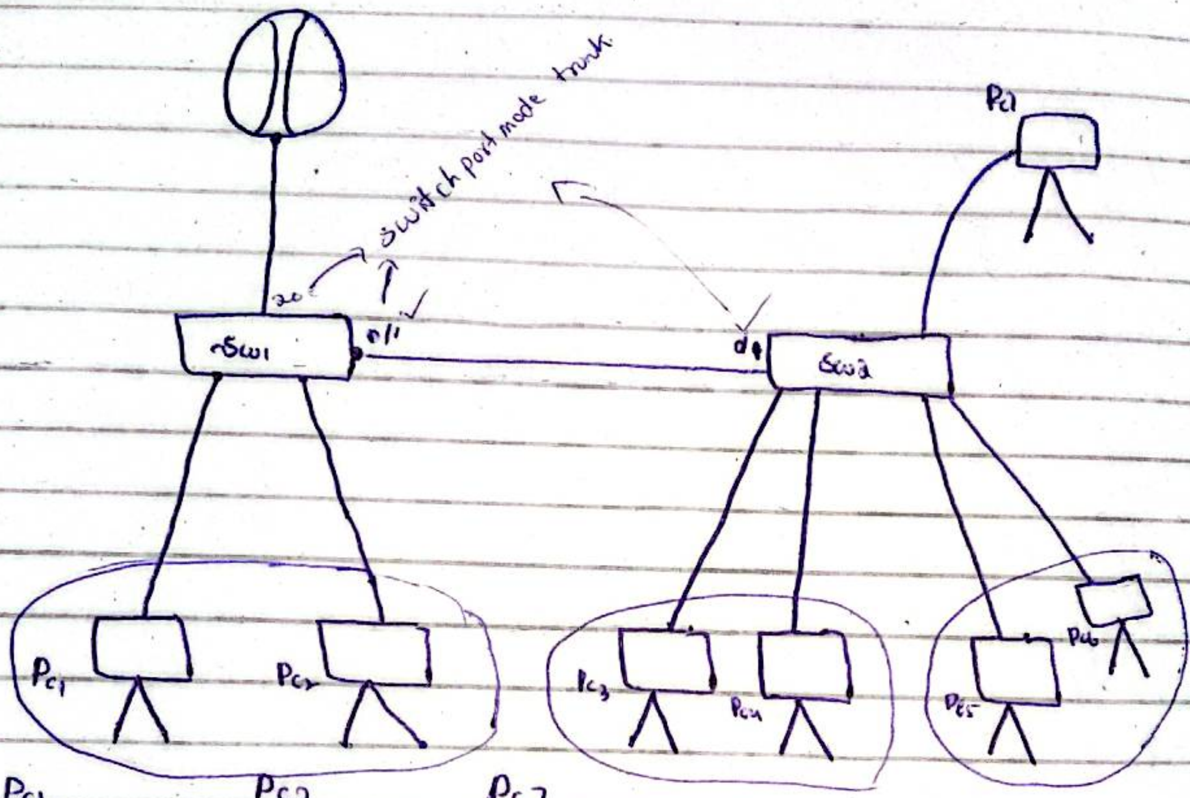
no shut.

exit

Same for 0/1

⇒ InterVLAN Router on stick

it is a methodology, where we divide the router's Port then it's sub ports we will connect VLAN's as we know that, Router has less ports, 1, or 2, or 3. so if we will have more VLAN's then we will need more Routers ports.



Pc1 16.0.0.1

Pc2 16.0.0.2

Pc7 16.0.0.3

Getway: 16.0.0.40 ______ ______

Pc3 192.168.0.1

192.168.0.40

Pc4 192.168.0.2

______

~~Pc5~~ ~~192.168.0.3~~

Pc5 192.168.1.30

Pc6 192.168.1.32

Router on stick

⇒ Now we want that these differerd networks will communicate with eachother.
So for that we connect the router.
So click on router:
click on cli mode.
→ no
→ enable
→ config t
→ Interface fastEthernet 0/0
→ No shut
→ exit.
→ Interface fastEthernet 0/0.1 // Now we are in router 0/0.1 port (subport)
→ encapsulation dot1Q 10 // Now this Port is dedicated to VLAN 10.
its Protocol.

→ ip add 16.0.0.40 255.0.0.0
→ no shut.
→ exit.
→ Now we are going to create another port. For that again
⇒ Interface FastEthernet 0/0.2
→ encapsulation dot1Q 20.
→ ip add 192.168.0.40 255.255.255.0.
→ exit.

⇒ Interface fastethernet 0/0.3
→ encapsulation dot1Q 30
→ ip add 192.168.1.40 255.255.255.0
→ exit

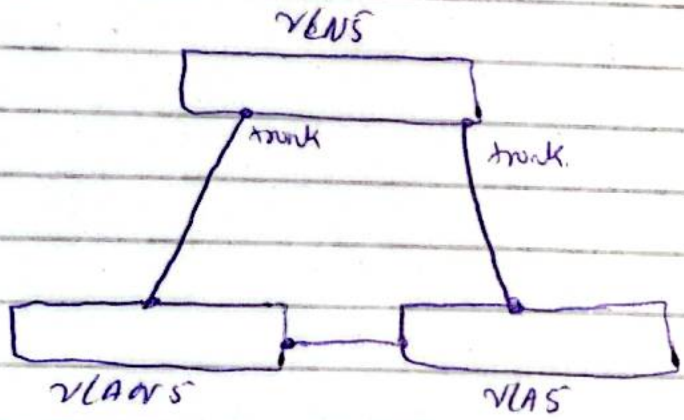
⇒ ~~VTP~~ (VLAN Trunking Protocol):-

But Before this topic there is another topic which is VLAN. and InterVLAN.

Def VTP:- To replicate the vlans from one switch to other switches

Note:- What VTP does? when we create the VLAN on just one switch, it will replic to all others ~~HA~~ swithes.

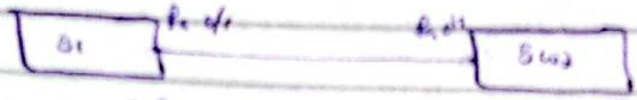
Means, when we create More VLAN's then we have to make each switchports as trunk.



⇒ we can made it easily without VTP when we have 2 or 3 switches but when we have 20, 30, 40, around 50 switches so due to this may be we Make some Mistakes. Means if we have more VLANS then obviously chance of human error.

⇒ So what VTP does, when we create VLAN on one switch it will replicate on all other switches.

So lets configure in Packet tracer



Means if we want to create VLAN in one Sw, then it will replicate to other switches.

So before configure the VTP, we have to made ports as trunk.

On Sw1

```
en
conf t
int. fa 0/1
Switchport mode trunk.
```

⇒ Note: for fast working ports enable use

```
en
config t
spanning-tree mode rapid-pvst
```

Note 2 when we do trunk on one side it will automatically does trunk on other side.

Note 3: How we can identify that our port is trunked:. for this write command:

→ show interfaces trunk.

→ so for Now configuring the VTP.
→ en
→ config t
→ vtp domain networkings → we can assign any thing to them.

→ Note: by default VTP domain Name is Null.
After Applying this command it will show VTP domain name from Null to networkings.

→ exit.
→ show vtp status. // this command will show vtp domain name.

⇒ Note: we learn VTP because if we assign/or Make VLAN on one Switch it will make that VLAN to All.

⇒ Note for that ~~VLAN~~ VTP domain Name should be same for all.

⇒ Sir Faiz Lectures.

ex:- we are working on large projects and want to creates huge prototypes. at that time we have more than one switches. and then when more than one switches exist, and we are creating the multiple VLANs., More than 20 or 30 VLAN so for those VLANs Numbers, VLANs Ids and their Names are diffigut to remember. means its configuration becomes difficult.

ex:- We are working on 4 switches and we want to create multiple vlans on one switch.

if I have create 4 vlan's on one switch then same 4 ~~VL~~ VLAN's we have to create for other switches., with same said, name etc. so by Avoiding this efforts we use VTP.

Note: We just configure VLAN's on just one, switch for other switches it will do it own self. automatically. all switches will sink to that particular switch.

So that means there will be no need for seprate configuration.

⇒ Modes of the VTP's:
1. VTP Mode Server
2. VTP Mode transparent
3. VTP Mode client.

⇒ Note1. VTP Mode server is our Main switch, at this swith we will do VTP Mode server.

Note2: Every switch domain name must be same. and, first switch mode we will made as a server.

⇒ Note3: Transparent switch will not sink with server's switch. we will make transparent if it will need.

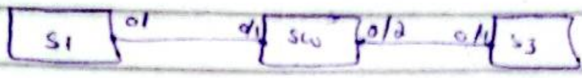
⇒ Note-4. Inside the transparent switch we can do addition, Modification, delation and If we have Made switch mode as a server. then where you can as addition, Modification and delation.

But those switches whose Mode is client, then you could not, add, remove or modification.

so this is the consept of VTP.

So for configuration:

First you have to take switches



step1 Now we have to change all ports mode as trunk.

→ So After trunking: we will made first ~~port~~ switch as a server.

and All the three switches VTP domain name we will make constant.

→ we will made first switch as a server and other as a client.

On SW 1

→ en

→ config t

→ vtp domain ~~name~~ faiz.

→ vtp Mode server //always remember that swith mode is server.

→

→ exit.

on switch 2

en
config t
vtp domain ~~name~~ faiz
vtp mode client. } same
exit.

Show vTP status.

Now on switch3.
en
config t
vTP domain faiz
vTP Mode client.
exit

Show vTP status.
exit.

Now what ever vlans I will create on the first switch ~~of~~ it will create on all other switches which are client Now, and first swith is server.

⇒ Now on first switch (server):

en
config t
VLAN 10
Name marketing
VLAN 20
Name hr
VLAN 30
name PP.
exit
exit
show VLAN

VTP status.

Now come on second switch which is client.
just write.
Show VLAN. // it will show that all VLANs are appended on others.

show status.

Now on third switch.
Show VLAN

so this is how VTP works